

MOST ASKED 30 CODING INTERVIEW QUESTIONS

AND FULL DETAILED 14-WEEK DSA ROADMAP

SECTION 1: MOST ASKED 30 INTERVIEW QUESTIONS (WITH LINKS)

1. Two Sum – <https://leetcode.com/problems/two-sum/>
2. Best Time to Buy and Sell Stock – <https://leetcode.com/problems/best-time-to-buy-and-sell-stock/>
3. Product of Array Except Self – <https://leetcode.com/problems/product-of-array-except-self/>
4. Maximum Subarray – <https://leetcode.com/problems/maximum-subarray/>
5. Longest Substring Without Repeating Characters – <https://leetcode.com/problems/longest-substring-without-repeating-characters/>
6. Longest Repeating Character Replacement – <https://leetcode.com/problems/longest-repeating-character-replacement/>
7. Binary Search – <https://leetcode.com/problems/binary-search/>
8. Search in Rotated Sorted Array – <https://leetcode.com/problems/search-in-rotated-sorted-array/>
9. Find First and Last Position of Element – <https://leetcode.com/problems/find-first-and-last-position-of-element-in-sorted-array/>
10. Merge Intervals – <https://leetcode.com/problems/merge-intervals/>
11. Insert Interval – <https://leetcode.com/problems/insert-interval/>
12. Non-overlapping Intervals – <https://leetcode.com/problems/non-overlapping-intervals/>
13. Reverse Linked List – <https://leetcode.com/problems/reverse-linked-list/>
14. Merge Two Sorted Lists – <https://leetcode.com/problems/merge-two-sorted-lists/>
15. Linked List Cycle – <https://leetcode.com/problems/linked-list-cycle/>
16. Valid Parentheses – <https://leetcode.com/problems/valid-parentheses/>
17. Min Stack – <https://leetcode.com/problems/min-stack/>
18. Daily Temperatures – <https://leetcode.com/problems/daily-temperatures/>
19. Binary Tree Level Order Traversal – <https://leetcode.com/problems/binary-tree-level-order-traversal/>
20. Maximum Depth of Binary Tree – <https://leetcode.com/problems/maximum-depth-of-binary-tree/>
21. Lowest Common Ancestor of a BST – <https://leetcode.com/problems/lowest-common-ancestor-of-a-binary-search-tree/>
22. Path Sum – <https://leetcode.com/problems/path-sum/>
23. Subsets – <https://leetcode.com/problems/subsets/>
24. Permutations – <https://leetcode.com/problems/permutations/>

- 25. Combination Sum – <https://leetcode.com/problems/combination-sum/>
- 26. Climbing Stairs – <https://leetcode.com/problems/climbing-stairs/>
- 27. House Robber – <https://leetcode.com/problems/house-robber/>
- 28. Longest Increasing Subsequence –
<https://leetcode.com/problems/longest-increasing-subsequence/>
- 29. Coin Change – <https://leetcode.com/problems/coin-change/>
- 30. Number of Islands – <https://leetcode.com/problems/number-of-islands/>

SECTION 2: FULL DETAILED 14-WEEK DSA ROADMAP

Week 1 – Foundations

- Time & Space Complexity, Big-O, loop analysis
- Array basics, traversal, max/min, reverse array
- Understand brute force vs optimized approach

Week 2 – Arrays & Hashing

- HashMap & HashSet usage
- Prefix sums, frequency counting
- Solve Two Sum, Stock Buy Sell, Product Except Self

Week 3 – Two Pointers & Sliding Window

- Left-right pointer patterns
- Fixed & variable window technique
- String and subarray problems

Week 4 – Stack & Queue

- Stack operations & use cases
- Monotonic stack concept
- Valid Parentheses, Daily Temperatures

Week 5 – Linked List

- Insert, delete, reverse linked list
- Fast & slow pointer technique
- Cycle detection and merge lists

Week 6 – Binary Search & Intervals

- Binary search on index & answer
- Sorting and custom comparators
- Merge and insert intervals

Week 7 – Recursion & Backtracking

- Recursion tree & base cases
- Backtracking template
- Subsets, permutations, combinations

Week 8 – Trees Basics

- DFS and BFS traversal
- Recursive tree problems
- Height, depth, traversal problems

Week 9 – Advanced Trees & BST

- BST properties
- LCA and path-based recursion
- Validate BST, Path Sum

Week 10 – Graphs

- Graph representation
- DFS, BFS, connected components
- Grid problems like Number of Islands

Week 11 – Dynamic Programming (1D)

- Overlapping subproblems
- Memoization vs tabulation
- Climbing Stairs, House Robber

Week 12 – Dynamic Programming (Advanced)

- State definition & transitions
- LIS and Coin Change
- 2D DP patterns

Week 13 – Mixed Revision

- Timed problem solving
- Pattern recognition practice
- Mock interview simulations

Week 14 – Interview Mode

- Revise all 30 core questions
- Explain solutions out loud
- Final mock interviews